

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (S,FE)/S1 (PT)(S) June 2024 (2019 Scheme)/S3 (WP)(R) December 2023 Examination

Course Code: EET203**Course Name: MEASUREMENTS AND INSTRUMENTATION**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions. Each question carries 3 marks*

Marks

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| 1 | Draw the block diagram of a typical measurement system and indicate the functional elements. | (3) |
| 2 | Illustrate the classification of different types of errors in measurements. | (3) |
| 3 | What is meant by creeping? What are the causes of creeping and how it can be eliminated? | (3) |
| 4 | Define the terms related to instrument transformer:
i) Transformation ratio ii) Turns Ratio iii) Nominal Ratio iv) Burden | (3) |
| 5 | Discuss the fall of potential method for earth resistance measurement. | (3) |
| 6 | With the help of diagram indicate the calibration of an ammeter using dc potentiometer. | (3) |
| 7 | Explain the measurement of flux density in a ring specimen. | (3) |
| 8 | With help of diagram explain the working of photo voltaic cell. | (3) |
| 9 | Illustrate the working of an ultrasonic flow transducer. | (3) |
| 10 | Discuss the working of a piezoelectric transducer. | (3) |

PART B*Answer any one full question from each module. Each question carries 14 marks***Module 1**

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| 11 | a | Illustrate different torques acting on moving system of an indicating instrument with help of neat diagram. | (7) |
| 11 | b | With the help of neat diagrams, explain the construction of attraction type moving iron instrument. Derive the expression for deflecting torque. | (7) |
| 12 | a | Discuss how the range of an instrument can be extended using shunts and multipliers. | (7) |

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- 12 b The inductance, L of a spring controlled moving iron meter is given by the following expression, $L = 1 + 0.0716 \theta - 0.0114\theta^2$ milli Henry where θ is deflection in radians. Control torque constant is $0.57 \times 10^{-3} \text{ Nm/rad}$. Calculate
- i) The current for full scale deflection of 120° . (4 marks)
- ii) Current for one half of full-scale deflection. (3 marks)

Module 2

- 13 a Discuss the two-wattmeter method for three phase power measurement with help of phasor diagram. Derive the expression for finding power factor of the load from wattmeter readings. (7)
- 13 b Explain the construction and working principle of an induction type energy meter. Derive the expression for driving torque. (7)
- 14 Derive the expression for transformation ratio and phase angle error of a current transformer using its equivalent circuit and phasor diagram. (14)

Module 3

- 15 a With neat circuit and phasor diagram explain how unknown capacitance is measured using Schering Bridge. (7)
- 15 b The arms of a four-arm bridge supplied with sinusoidal voltage, have the following values, (7)
- Arm ab- A resistance 200Ω in parallel with capacitor $1 \mu\text{F}$.
Arm bc- 400Ω resistance, Arm cd- 1000Ω resistance
Arm da- R_2 in series with a $2 \mu\text{F}$ capacitor. Determine R_2 & frequency at which bridge will balance?
- 16 a Which bridge can be used to measure low resistances? Draw the diagram of this bridge and show that measurement of resistance using this bridge is independent of connecting lead resistance. (7)
- 16 b Draw the circuit diagram of a bridge used for frequency measurement. Also derive the equation for frequency when the bridge is balanced. (7)

Module 4

- 17 a Illustrate the determination BH-curve & hysteresis loop of a magnetic material by using step by step method (7)
- 17 b With neat sketch, explain the Lloyd fisher square method for measurement of core loss in a magnetic material. (7)

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- 18 a Describe method of reversal for the determination of hysteresis loop of a magnetic material. (7)
- 18 b Compare temperature measurement using RTD, thermocouple and thermistors. (7)

Module 5

- 19 a With the help of neat sketch explain the working of LVDT. Also draw its characteristics (7)
- 19 b Draw the block diagram and explain the working of digital Storage oscilloscope. (7)
- 20 a With neat block diagram explain the working of a CRO. (7)
- 20 b How frequency and phase is measured using Lissajous pattern. (7)